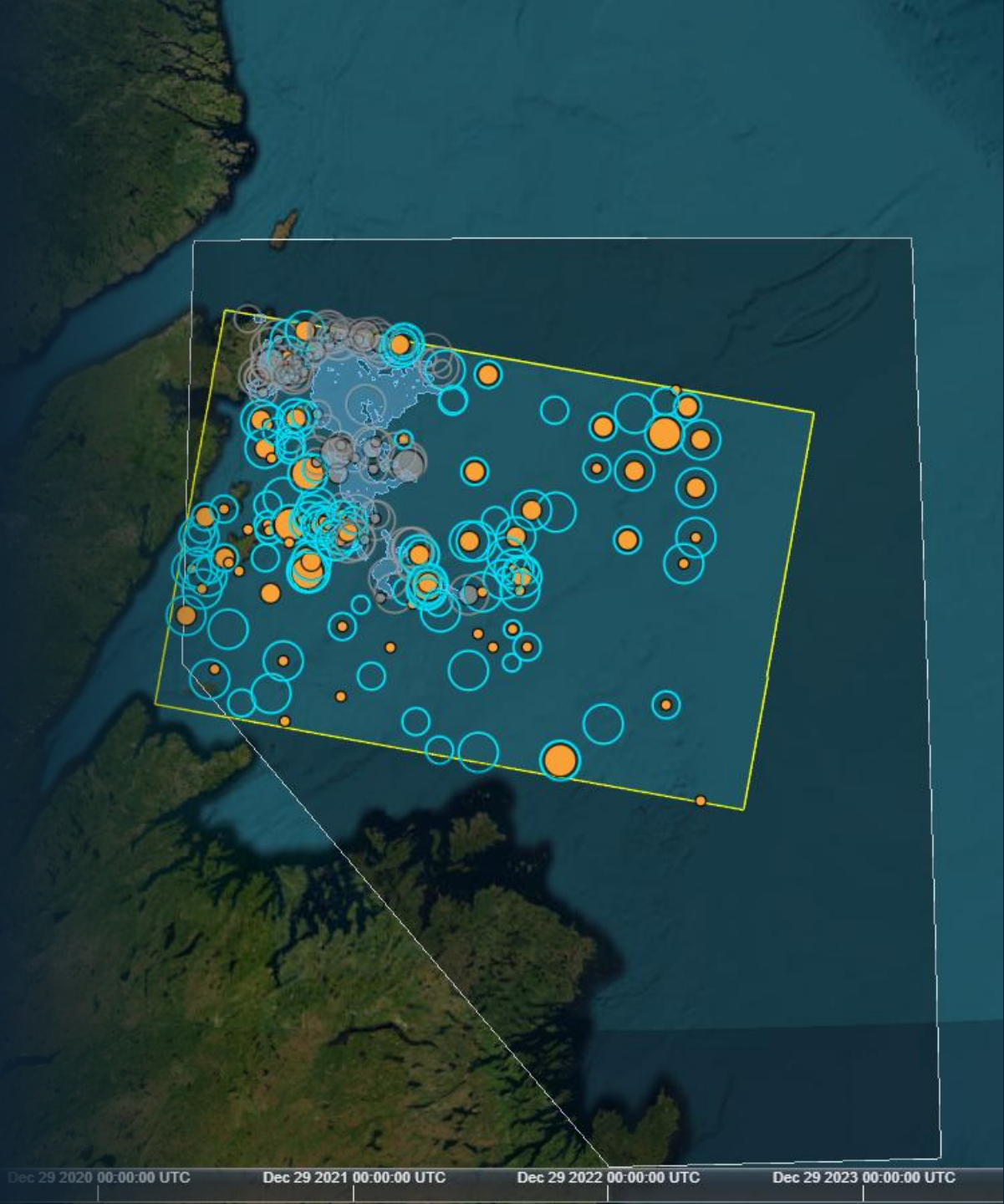
PROJECT BRIEF · SAR + FOUNDATION MODEL

Iceberg Alley SAR

Finding icebergs off Newfoundland in satellite radar with THOR, a SAR foundation model, benchmarked against a classical detector and shown on a live globe.

Natali Kovalski · October 2026



THE PROBLEM

Every spring, icebergs drift into busy water

01

The hazard

Greenland bergs ride the Labrador Current down "Iceberg Alley", past shipping lanes, fishing grounds and offshore platforms.

02

The gap

Aircraft reconnaissance is costly and limited by weather, and this coast is often fogged in.

03

The idea

Radar sees through cloud and darkness. A foundation model pretrained on Sentinel-1 can learn to find bergs from a few hundred labels.

HEADLINE RESULT

84%

of the icebergs Ice Patrol analysts marked on four held-out passes were found by THOR: **431 of 512**. A classical CA-CFAR found 380, with as many detections.

April 2019 · same masks, labels, matching and detection count · large bergs: 11 of 14 vs. 4

APPROACH

Six steps from raw scene to globe

01

Search

ASF search for HH+HV scenes over the AOI

02

Calibrate

HyP3 terrain correction to 10 m, sigma0 for THOR

03

Mask

Land plus 500 m from OSM; pack ice from HV texture

04

Detect

THOR: frozen encoder + 769-parameter head

05

Baseline

CA-CFAR on HV, same masks, same output

06

Publish

GeoJSON and CZML, published to a live web viewer

THOR and the CFAR share the masks and the output schema, so both run through one validation and one viewer.

SAR AND ML IN ONE SLIDE

Four ideas behind the detector

Backscatter

Calm water reflects radar away and looks dark. An iceberg's facets send energy back and look bright.

Polarization

Sea clutter is far weaker in cross-pol (HV), so bergs stand out there. HH is the second check.

Speckle

Grainy noise in every SAR image. Its strength (ENL) is measured per scene and sets the threshold.

Foundation model

A network pretrained on huge amounts of unlabelled imagery. A tiny head on its features can learn a new task from a few hundred labels.

THE DETECTOR

A frozen foundation model, a tiny head

01

Frozen encoder

THOR Tiny (FM4CS, Apache 2.0), pretrained on Sentinel-1 sigma0. It takes HH/HV natively and is never fine-tuned.

02

769-parameter head

A linear layer on cached features turns them into an iceberg heatmap. Training takes minutes on a laptop GPU.

03

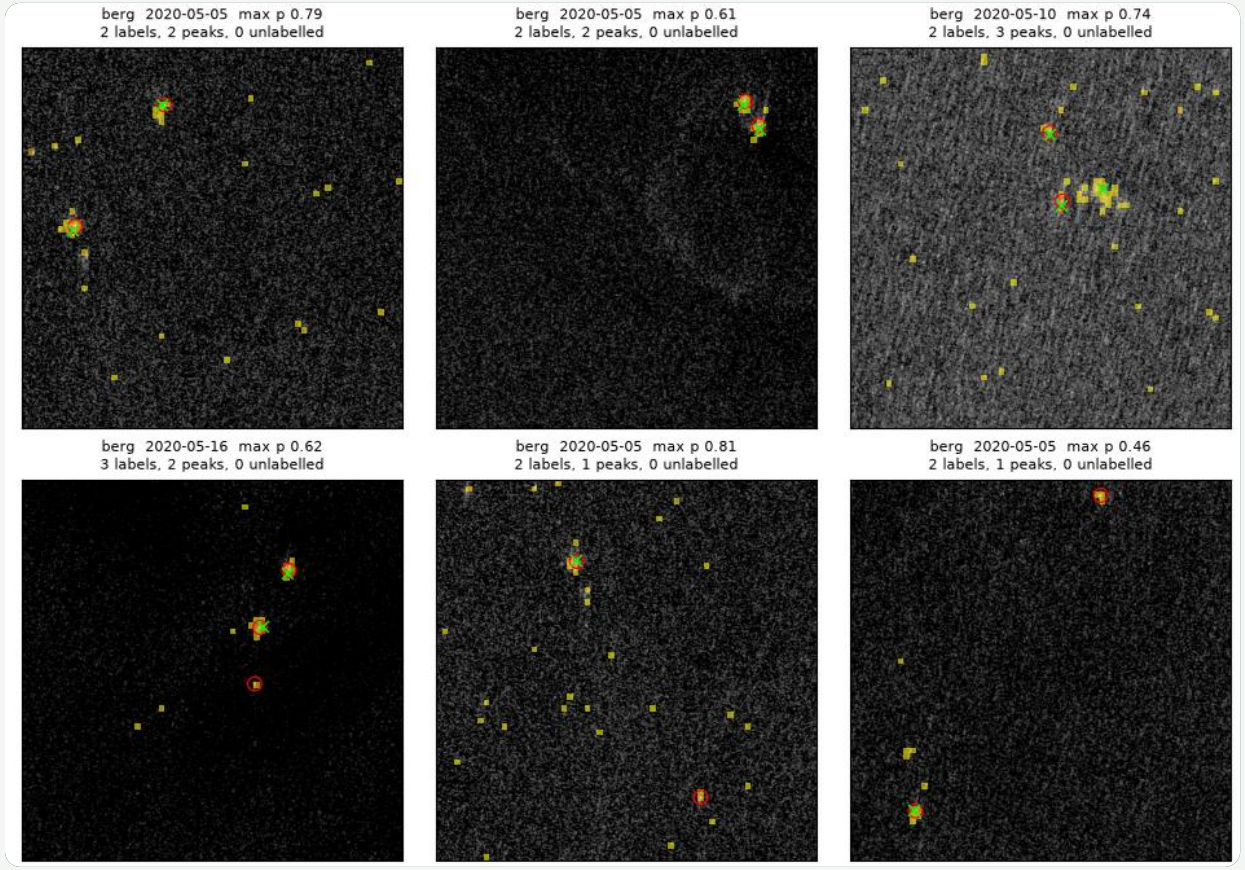
Same masks

Peaks must fall on the classical masked raster. Output is the CFAR's GeoJSON schema, plus a score.

TRAINING

Learned from Ice Patrol labels

- **Labels:** IIP analysts' berg positions on the same pass
- **Data:** 723 crops from 20 passes, 2019–2020
- **Held out:** all of April 2019, for the test
- **Detections:** heatmap peaks scoring ≥ 0.35

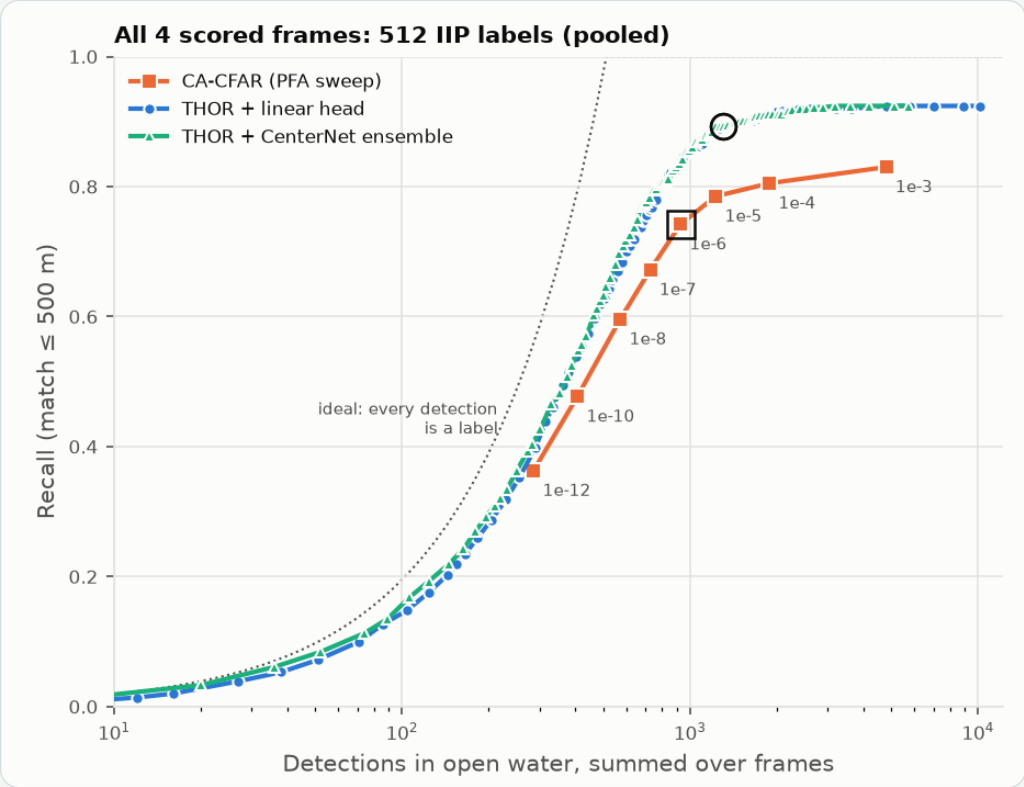


THOR heatmaps on 2020 validation crops: IIP labels (red circles), THOR peaks (green ×).

THOR VS. CA-CFAR

10 points more bergs at the same count

Recall	CFAR	THOR
4 passes, 512 bergs	0.74	0.84
2019-04-29 only	0.74	0.89
Medium, 61–120 m	0.76	0.87
Large, over 120 m	4 of 14	11 of 14
Small, 15–60 m	0.71	0.71

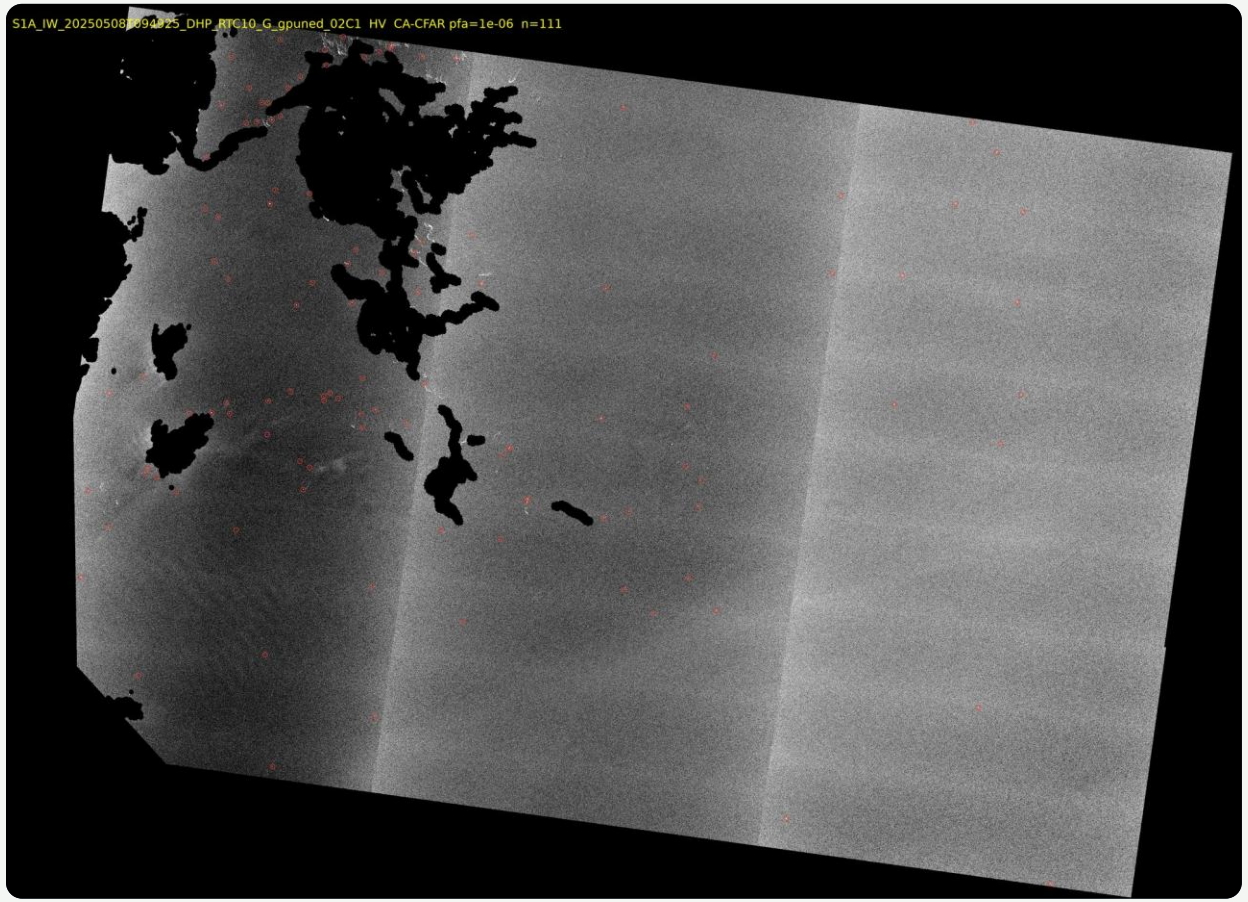


Recall vs. open-water detections, 4 held-out passes. Square: CFAR at PFA 10^{-6}.

THE BASELINE

CA-CFAR: the classical detector

- **Threshold** from a 10^{-6} false-alarm rate, speckle measured per scene
- **Filters:** size, ice strips over 300 m, HH co-pol check
- **Strength:** no training, runs on a CPU
- **Weakness:** the strip rule drops large bergs



HV, 2025-05-08. Land and pack ice masked (black), 111 CFAR detections circled.

VALIDATION

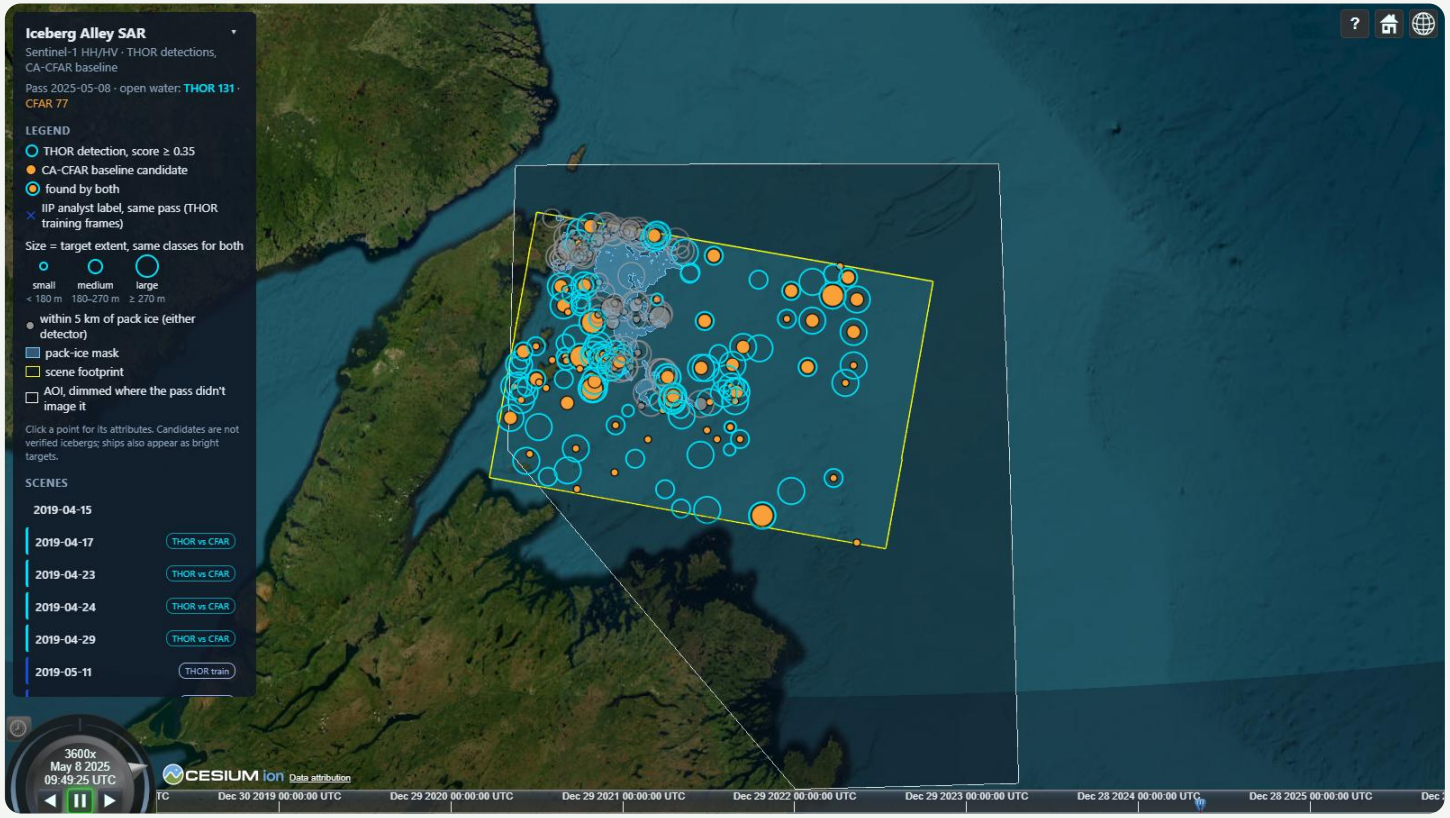
Every ground-truth source has a catch

Source	What it gives	Outcome
NAIS iceberg chart	Counts per 1° square	Within 2–4×, but opposite errors cancel
IIP aircraft sightings	Positions 5–7 h after the pass	Inconclusive: drift unresolved
Sentinel-2 optical	Bright objects at 10 m	Loose sea ice in April, bergs inshore in June
IIP labels, same pass	Positions on the same image	Train and test sets: no drift

Where truth comes hours later, the code estimates a common drift offset and checks it against chance. If the peak is weak, it reports "drift unresolved" instead of a score.

VIEWER

Every candidate on a globe, by date



- 32 scenes (28 passes), 2019–2026, live on GitHub Pages
- Cyan rings: THOR. Orange dots: CA-CFAR baseline
- Test passes open their THOR vs. CFAR chart
- Grey: within 5 km of pack ice, likely floes

LIMITS AND NEXT STEPS

What the numbers don't show yet

Limits

- **Precision unknown** for both: unlabelled detections may be bergs, boats or clutter
- **One month of one year:** April 2019, 512 bergs
- **THOR needs the pipeline:** the pack-ice mask, and a GPU
- **Small bergs:** a tie at 0.71

Next

- **AIS ship positions** for the test passes, to measure precision
- **THOR at 4×4 patches** for small bergs
- **THOR in the one-command run** and as the viewer's main layer
- **More years** and near-range passes

BUILD

Two commands, end to end

```
python -m iceberg_sar.cli run --start 2025-05-01 --end 2025-05-15
python -m iceberg_sar.cli thor-detect data/raw/<scene>
```

Processing

Python 3.11, NumPy, SciPy, rasterio, GeoPandas

Model

THOR Tiny via TerraTorch, PyTorch, CUDA GPU

Viewer

Vue 3, CesiumJS, CZML, deployed by GitHub Actions

Quality

153 pytest tests, all parameters in config.yaml

run searches, orders HyP3 processing, masks and runs the baseline; thor-detect adds THOR. Then publish and git push update the live viewer.

THANK YOU

Questions?

Natali Kovalski · github.com/natali-kovalski/iceberg-alley-sar

Live demo: natali-kovalski.github.io/iceberg-alley-sar. Contains modified Copernicus Sentinel data (2019–2026). Iceberg data: International Ice Patrol (NSIDC G00807) and the North American Ice Service. THOR: FM4CS (Apache 2.0); Forgaard et al., IGARSS 2026. Land polygons © OpenStreetMap contributors.